Detecting Voice Forgeries Based on Neural Audio Codecs — CodecFake

Wu, Tseng, and Lee (2024), in their paper *CodecFake: Enhancing Anti-Spoofing Models Against Deepfake Audios from Codec-Based Speech Synthesis Systems*, investigate the ability of existing voice forgery detection systems to cope with speech created by means of speech synthesis systems based on Neural Audio Codecs. These systems use discrete audio representations as part of the speech-generation process, and enable the production of high-quality synthetic speech, including the imitation of speakers who were not included in training. The researchers point to a gap between the progress in speech-generation systems and the generalization ability of Anti-Spoofing models trained on existing datasets, in particular ASVspoof 2019.

In order to examine the problem, the researchers presented the CodecFake dataset, built by means of six types of Neural Audio Codec Frameworks including 15 pre-trained models. Unlike full speech generation by means of Codec Language Models, the dataset-creation process is based on Codec Resynthesis: a real speech recording is encoded into a sequence of discrete codes by means of a Codec Encoder, and is then reconstructed by means of a Codec Decoder. This approach makes it possible to focus on the acoustic features related to the encoding and decoding process, while reducing the dependence on differences between speech-generation models. The dataset is built from VCTK, and the researchers used a separate split by speakers between the training, validation and testing sets, in order to reduce the possibility that the models would learn to identify specific speakers instead of forgery features.

For the evaluation, the researchers used the AASIST-L architecture and compared models trained on ASVspoof 2019 Logical Access to models trained on different CodecFake subsets. In Cross-Testing experiments, each model was trained on examples reconstructed by means of a particular Codec and on the corresponding source recordings, and was then tested on examples created by means of all 15 Codec models. The results showed that models trained on ASVspoof 2019 struggled to detect a significant portion of the reconstructed examples, and in most of the subsets Equal Error Rate (EER) rates above 30% were obtained. In contrast, training on a particular Codec usually yielded better results when the testing was carried out on the same Codec or on models from the same family. However, in some cases significant gaps were observed between performance on known attacks and performance on other Codec types, which emphasizes the challenge of generalization between different audio-generation mechanisms.

In order to examine whether a dataset based on Resynthesis can also help detect speech created by means of full synthesis systems, the researchers evaluated the models on examples from VALL-E, from VALL-E X and from SpeechX. On the VALL-E system, an AASIST-L model trained on ASVspoof 2019 achieved an EER of 13.91%, while a model trained on CodecFake Subset E, based on EnCodec, achieved an EER of 0.40%. This result demonstrates that Codec Resynthesis data can help detect speech created by means of a full Codec-Based Speech Synthesis system, even when the speech-generation model itself is not part of the dataset-building process. Improvements were also observed on VALL-E X and on SpeechX, but the researchers note that the number of evaluation examples collected from the demo pages is small, and in particular the statistical significance of the improvement for SpeechX could not be verified. In addition, the performance varied considerably depending on the Codec subset chosen for training, and therefore it should not be concluded that every type of Codec provides similar generalization ability.

From a research perspective, the central contribution of CodecFake is in demonstrating the difficulty of Anti-Spoofing systems in generalizing to synthesis technologies based on Neural Audio Codecs, and in presenting a method for building training data intended to reduce this gap. However, the study focuses on a single central detection architecture, AASIST-L, and on a limited number of full synthesis systems. In addition, the contribution of joint training on all the subsets was not systematically examined, and generalization ability to every future Codec-Based system was not proven. Therefore, the results should be interpreted as evidence of the possible usefulness of Codec Resynthesis data, and not as proof of a universal solution for voice forgery detection.

For the present study, the CodecFake paper is especially relevant because part of the project's recording dataset originates from CodecFake, alongside examples from ASVspoof 5 and from M-AILABS. The paper emphasizes the need to evaluate models not only by means of overall EER, but also while examining the generalization ability between data sources and attack types. Accordingly, approaches based on classical acoustic features, on AASIST, and on pre-trained speech representations such as wav2vec 2.0 and WavLM can be compared, while maintaining a uniform evaluation protocol and preventing information leakage. In addition, it should be examined whether the models learn general forgery features or rely on features unique to the data source. This study thus reinforces the importance of Cross-Dataset Generalization as a central component in evaluating voice forgery detection systems.

Bibliographic source:

Wu, H., Tseng, Y., & Lee, H.-y. (2024). *CodecFake: Enhancing Anti-Spoofing Models Against Deepfake Audios from Codec-Based Speech Synthesis Systems*. arXiv:2406.07237. https://arxiv.org/abs/2406.07237